

AJITESH SHUKLA

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EDUCATION

Vellore Institute of Technology, Bhopal, MP
B.Tech CSE (Spec. in AI & ML) — CGPA: 9.08/10

Expected May 2027

TECHNICAL SKILLS

- **Languages:** C++, Python, SQL
- **Libraries:** Pandas, NumPy, Scikit-Learn, TensorFlow, Keras, Matplotlib, Seaborn, PyTorch
- **Tools:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab
- **Concepts:** Data Structures (DSA), OOP, DBMS, Machine Learning, Deep Learning, Computer Vision

CERTIFICATIONS

- **AWS Certified AI Practitioner** — View Certificate
- **Oracle Cloud Infrastructure 2025 Certified Data Science Professional** — View Certificate
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** — View Certificate
- **Applied Machine Learning in Python** (University of Michigan) — View Certificate

PROJECTS

Delhi Air (CPCB AQI Dashboard & LSTM Forecaster) — Source Code May 2026 – Present

- Addressed a 44% mean error and low 0.401 correlation in standard satellite-estimated AQI feeds by engineering an end-to-end Python engine that streams official ground-truth CPCB data via paginated REST APIs and applies true CPCB rolling-window sub-indices.
- Architected a multi-input hybrid Conv1D-LSTM model in PyTorch/Keras using 168-hour historical pollutant sequences, future weather forecasts, and dynamic station embeddings to predict 24-hour PM2.5 residuals with Huber loss and gap-aware sequence filtering.
- Achieved low latency model inference speed and deployed a responsive Streamlit dashboard featuring automated health-risk indicators, interactive spatial visualizations, and an automated GitHub Actions ETL pipeline managing dual Parquet stores.

Image Dehazing using Autoencoders — Source Code Nov 2025 – Present

- Architected a U-Net based deep autoencoder in PyTorch to eliminate atmospheric haze and restore structural clarity from degraded images.
- Integrated residual learning blocks and spatial attention mechanisms to capture fine edge details and prevent feature degradation across deep layers.
- Optimized training using a composite loss function blending Perceptual (VGG-16) and SSIM loss to improve color fidelity and structural reconstruction.
- Built a synthetic haze pipeline across 10,000+ image pairs, achieving 28.5 dB PSNR and 0.92 SSIM on benchmark validation datasets with sub-50ms inference time.

Tsunami Prediction — Source Code Jan 2026 – Present

- Engineered a Random Forest early warning system to predict tsunami probabilities from real-time seismic parameters.
- Resolved 95% class imbalance using balanced class weights and Stratified K-Fold cross-validation, optimizing ROC-AUC to minimize false negatives.
- Performed feature engineering through logarithmic depth scaling and energy interactions, deploying the predictive pipeline via Streamlit.

EXPERIENCE

MPOnline Limited — Software Engineering Intern May 2026 – July 2026

- Completed a virtual internship on core software engineering principles and built hands-on projects.
- Participated in technical viva evaluations, real-world problem-solving discussions, and project architecture sessions.

ACHIEVEMENTS

- **Health Hackathon 2026:** Finalist — placed in top 40 out of 600+ teams in hackathon organized by VIT Bhopal and Johns Hopkins University.
- **Flipkart GRiD 8.0:** Semi-Finalist — Reached Round 3 in Flipkart's flagship national engineering challenge.
- Actively strengthening Data Structures and Algorithms proficiency through continuous practice on **LeetCode** and **GeeksforGeeks**.
- **Dr. G. Viswanathan Challenge:** Completed 100 Days of Code, strengthening core logic.